

microSD™ Card Connectors

DM3 Series



■ Features

◆ Common to the entire Series

1. Extremely small in size

Small external dimensions and the above-the-board height make the connectors the smallest on the market.

2. Reverse card insertion protection

Unique card slot design (patented) protects the connector from damage when the card is attempted to be inserted in reverse, allowing it to re-inserted correctly.

3. Effective ground and shield configuration

4-connection points of the metal cover to the printed circuit board assures secure connection of the ground circuit and provides EMI protection.

4. Rigid and strong construction

Despite its small size, high-strength materials used in the connectors produced a strong and rigid structure.

5. Card detection switch

The card detection switch is Normally Open

◆ DM3AT and DM3BT (Push - Push, with ejection mechanism)

• Card fall-out prevention

Built-in card tray and the unique push insertion-push ejection mechanism (patented) prevent accidental card ejection or fall-out.

Despite its small size the connectors will eject the card to a distance of 4.0mm, allowing easy hold and removal of the card.

• Exposed termination leads

Easy inspection and rework of the solder termination joints.

◆ DM3CS (Hinge, Push-Pull, manual, without ejection mechanism)

• Simple and reliable card insertion

Hinged metal cover provides location and guides the card during the insertion / removal. Closing of the cover confirms the electrical and mechanical connection with a tactile click sensation.

• Reliable contact with the card contact pads

Unique contact design and card slide action will clean the contact areas of the card.

• Accessible termination areas

Contact solder terminations may be inspected and reworked.

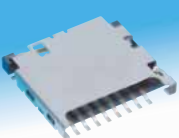
◆ DM3D (Push -Pull, manual, without ejection mechanism)

• Partial card insertion hold

Card will not fall-out even when it is not fully inserted. Full insertion and electrical / mechanical connection is confirmed with a distinct tactile feel.

• Accessible termination areas

An inner lead system that can be reworked is used in this design. Contact solder terminations may be inspected and reworked.

Card insertion-ejection	Series	Image	Page
Push-Push	DM3AT		2~4
	DM3BT		5~6
Hinge-manual insertion/ejection	DM3CS		7~8
Push-Pull manual insertion/ejection	DM3D		9~10

■Product Specifications (DM3 Series)

Ratings	Current rating : 0.5A Voltage rating : 125V AC	Operating temperature range : -25°C to +85°C (Note 1) Storage temperature range : -40°C to +85°C (Note 2)	Operating humidity range : RH 95% max. (No condensation)
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Item	Specification	Conditions
1. Insulation resistance	1000MΩ min. (Initial value)	Measure at 500V DC
2. Withstanding voltage	No flashover or insulation breakdown	500 V AC / 1 minute
3. Contact resistance	100mΩ max. (Initial value)	1mA
4. Vibration	No electrical discontinuity of 100 ns or longer No damage, cracks or parts dislocation.	Frequency : 10 to 55Hz, single amplitude of 0.75mm, 3 directions for 2 hours
5. Humidity	Contact resistance : 40mΩ max. (change from initial value) Insulation resistance : 100MΩ min. No damage, cracks or parts dislocation.	96 hours at of 40 ± 2°C, and humidity of 90 to 95%
6. Temperature cycle	Contact resistance : 40mΩ max. (change from initial value) Insulation resistance : 100MΩ min. No damage, cracks or parts dislocation.	-55°C → 5 to 35°C → +85°C → 5 to 35°C Times : 30 min. → 5 min. → 30 min. → 5 min. 5 cycles
7. Durability	Contact resistance : 40mΩ max. (change from initial value)	10,000 cycles, 400 to 600 cycles per hour (DM3AT, DM3B) 5,000 cycles, 400 to 600 cycles per hour (DM3C, DM3D)
8. Resistance to soldering heat	No deformation of components affecting performance.	Reflow : At the recommended temperature profile Manual soldering : 350°C for 3 seconds

Note 1 : Includes temperature rise caused by current flow.

Note 2 : The term "storage" refers to products stored for long period prior to mounting and use.

■Materials / Finish

DM3AT, DM3BT

Part	Material	Finish	Remarks
Insulator	LCP	Color : Black	UL94V-0
Contacts	Copper alloy	Contact area : Gold plated Lead area : Gold plated	_____
Guide cover	Stainless steel (DM3AT) Copper alloy (DM3BT)	Lead area : Gold plated	_____
Other components	Stainless steel (DM3AT, DM3BT) Piano wire (DM3BT)	_____ Nickel plated	_____

DM3CS, DM3D

Part	Material	Finish	Remarks
Insulator	LCP	Color : Black	UL94V-0
Contacts	Copper alloy	Contact area : Gold plated Lead area : Gold plated	_____
Guide cover	Stainless steel	_____ (DM3CS) Tin plated (DM3D)	_____

■Product Number Structure

Refer to the chart below when determining the product specifications from the product number.

Please select from the product numbers listed in this catalog when placing orders.

DM3 AT – SF – PEJM5

①

②

③

④

① Series name : DM3	③ Termination type : SF Right-angle SMT(Standard) DSF Right-angle SMT(Reverse)
② Connector type : AT Push-Push (ejection mechanism), Top board mounting (Standard) BT Push-Push (ejection mechanism), Bottom board mounting (Reverse) CS Hinge, Push-Pull (no ejection mechanism), Top board mounting (Standard) D Push-Pull (no ejection mechanism), Top board mounting (Standard) Number of contacts : 8	④ Card ejection code : PEJM5, PEJS (Push insert/push eject) None : Manual card insertion/ejection

Technical drawing of the HS5 card reader, showing dimensions in millimeters (mm).

Top View Dimensions:

- Overall width: 13.85
- Distance from left edge to center of card slot: 7.35
- Distance from center of card slot to right edge: 1.68
- Distance from left edge to center of card slot (alternative measurement): 3.2
- Distance from left edge to center of card slot (alternative measurement): 15.95
- Distance from center of card slot to right edge (alternative measurement): 15.95
- Distance from center of card slot to right edge (alternative measurement): 16.75; CARD OVER STROKE POSITION
- Distance from center of card slot to right edge (alternative measurement): 17.55; CARD LOCK POSITION
- Distance from center of card slot to right edge (alternative measurement): 21.55; CARD EJECT POSITION
- Distance from left edge to center of card slot (alternative measurement): 5.5
- Distance from center of card slot to right edge (alternative measurement): 11
- Distance from center of card slot to right edge (alternative measurement): 0.8

Side View Dimensions:

- Overall height: 15.95
- Distance from bottom edge to center of card slot: 15.95
- Distance from bottom edge to center of card slot (alternative measurement): 16.75; CARD OVER STROKE POSITION
- Distance from bottom edge to center of card slot (alternative measurement): 17.55; CARD LOCK POSITION
- Distance from bottom edge to center of card slot (alternative measurement): 21.55; CARD EJECT POSITION

Front View Dimensions:

- Overall width: 13.85
- Distance from left edge to center of card slot: 7.35
- Distance from center of card slot to right edge: 1.68
- Distance from left edge to center of card slot (alternative measurement): 3.2
- Distance from center of card slot to right edge (alternative measurement): 15.95
- Distance from center of card slot to right edge (alternative measurement): 16.75; CARD OVER STROKE POSITION
- Distance from center of card slot to right edge (alternative measurement): 17.55; CARD LOCK POSITION
- Distance from center of card slot to right edge (alternative measurement): 21.55; CARD EJECT POSITION
- Distance from left edge to center of card slot (alternative measurement): 5.5
- Distance from center of card slot to right edge (alternative measurement): 11
- Distance from center of card slot to right edge (alternative measurement): 0.8

Pin Headers:

- #1(DAT2)
- #2(CD/DAT3)
- #3(CMD)
- #4(VDD)
- #5(CLK)
- #6(VSS)
- #7(DAT0)
- #8(DAT1)

Other Labels:

- CARD DETECTION SWITCH(A)
- CARD DETECTION SWITCH(B)
- microSD CARD
- HS5
- 2.9 ± 0.15

All dimensions : mm

Technical drawing of a mechanical part with dimensions in mm. The drawing shows a cross-section of a component with various features like a hole, a slot, and a fillet. Dimensions are given in mm, with some values in parentheses indicating alternative or maximum/minimum values. Key dimensions include: overall width 14.05, overall height 15.5, hole diameter 10MIN, hole position 9.1, slot width 8.9MIN, slot position 9.25MAX, and fillet radius R0.15. Surface finish symbols are present on several surfaces.

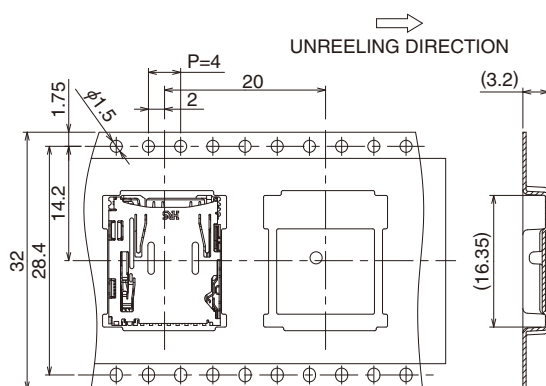
2

Card detection switch	
Without the card	Card inserted
Open (A) (B)	Closed (A) (B)

All dimensions : mm

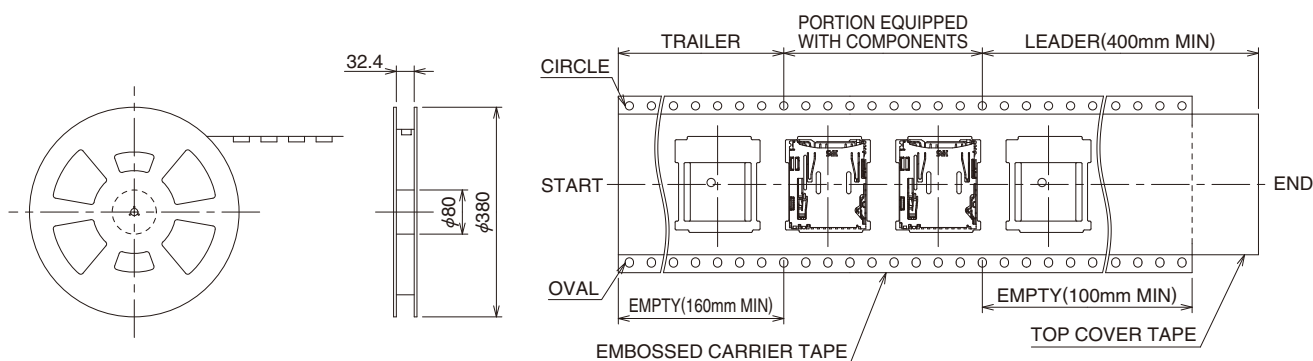
◆ Packaging Specifications

● Embossed carrier tape dimensions (1,500 pcs/reel)



All dimensions : mm

● Reel Dimensions



Technical drawing of the HS-01 module, showing top, side, and front views with dimensions and pin labels.

Pin Labels (Top View):

- #4(VDD)
- #5(CLK)
- #6(VSS)
- #7(DAT0)
- #8(DAT1)
- #1(DAT2)
- #2(CD/DAT3)
- #3(CMD)

Dimensions (mm):

- Top View: 13.85, (3.2), (6.5), 2.9±0.15
- Side View (Left): 2.38MAX, 1.83, (0.55)
- Side View (Right): 2.7MAX, 6.7MAX, 15.1, 15.45
- Front View: (21.55):CARD EJECT POSITION, (17.55):CARD LOCK POSITION, (16.75):CARD OVER STROKE POSITION, 16.05, 15.95, (15), (5.5), (11), 2.9±0.15

Other Labels:

- CARD DETECTION SWITCH(B)
- CARD DETECTION SWITCH(A)
- microSD CARD
- HS
- Q

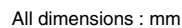
All dimensions : mm

2	Card detection switch	
	Without the card	Card inserted
	Open (A) (B)	Closed (A) (B)

4  No conductive traces.

HRS 5

● Embossed carrier tape dimensions (1,200 pcs/reel)



Technical drawing of the 35mm film magazine, showing three views: top, side, and front.

Top View: A circular view showing the internal frame with five sprocket holes. The outer diameter is $\phi 98.0$ mm. The inner diameter of the frame is $\phi 80$ mm. The distance between the sprocket holes is 32.4 mm.

Side View: A cross-sectional view showing the internal components. The total height is 32.4 mm. The distance from the top edge to the center of the sprocket holes is 80 mm. The distance from the center of the sprocket holes to the bottom edge is 98.0 mm.

Front View: A detailed view of the magazine's front face, showing the sprocket holes and the internal frame. The frame is divided into sections: CIRCLE, TRAILER, PORTION EQUIPPED WITH COMPONENTS, LEADER (400mm MIN), and END. The sprocket holes are arranged in a circle with a diameter of 160mm (EMPTY) and 100mm (EMPTY). The sprocket holes are labeled as EMBOSSED CARRIER TAPE and TOP COVER TAPE.

CARD DETECTION SWITCH ≤ 2
 13.8
 (6.9)
 (0.2)
 14.1
 (15.45): LOCK POSITION
 12
 13.9
 1.83
 0.1
 (0.93)
 (1.78)
 (5.5)
 (11)
 microSD CARD
 # 8(DAT1)
 # 7(DAT0)
 # 6(VSS)
 # 5(CLK)
 # 1(DAT2)
 # 2(CD/DAT3)
 # 3(CMD)
 # 4(VDD)
 All dimensions : mm

[illegible]

2

Card detection switch	
Without the card	Card inserted
Open	Closed

All dimensions : mm

- Embossed carrier tape dimensions (1,300 pcs/reel)



Technical drawing of the 35mm film magazine, showing three views: top, side, and front.

Top View: A circular view showing the internal layout of the film magazine. The outer diameter is $\phi 360$ mm. The inner diameter of the central hub is $\phi 80$ mm. The thickness of the magazine is 32.4 mm.

Side View: A cross-sectional view showing the internal components. The total length is 400 mm. The central section is labeled "PORTION EQUIPPED WITH COMPONENTS". The left section is labeled "TRAILER" and the right section is labeled "LEADER(400mm MIN)".

Front View: A detailed view of the front cover tape. The tape is labeled "EMBOSSD CARRIER TAPE". The central section is labeled "PORTION EQUIPPED WITH COMPONENTS". The left section is labeled "TRAILER" and the right section is labeled "LEADER(400mm MIN)". The tape is labeled "START" and "END". The tape is labeled "OVAL". The tape is labeled "EMPTY(160mm MIN)" and "EMPTY(100mm MIN)".

Figure 1: Dimensions of the module. The diagram shows the module's footprint and internal components. Key dimensions include:

- Overall width: 11.95 mm
- Overall height: 15.8 mm (CARD FULLY INSERTED)
- Card Detection Switch (B) width: 1.55 mm
- Card Detection Switch (A) width: 1.55 mm
- MicroSD Card slot width: 4.5 mm
- MicroSD Card slot height: 11 mm
- Pin 1 (DAT2) position: 11.45 mm from the right edge
- Pin 8 (DAT1) position: 11.45 mm from the left edge
- Pin 4 (VDD) position: 4.5 mm from the bottom center
- Pin 5 (CLK) position: 4.5 mm from the bottom center
- Pin 6 (VSS) position: 4.5 mm from the bottom center
- Pin 7 (DAT0) position: 4.5 mm from the bottom center
- Pin 3 (CMD) position: 4.5 mm from the bottom center
- Pin 2 (CD/DAT3) position: 4.5 mm from the bottom center
- Pin 1 (DAT2) position: 4.5 mm from the bottom center

Pin numbers 1 through 8 are labeled at the bottom, corresponding to DAT1, DAT0, VSS, CLK, VDD, CMD, CD/DAT3, and DAT2.

All dimensions : mm

[illegible]

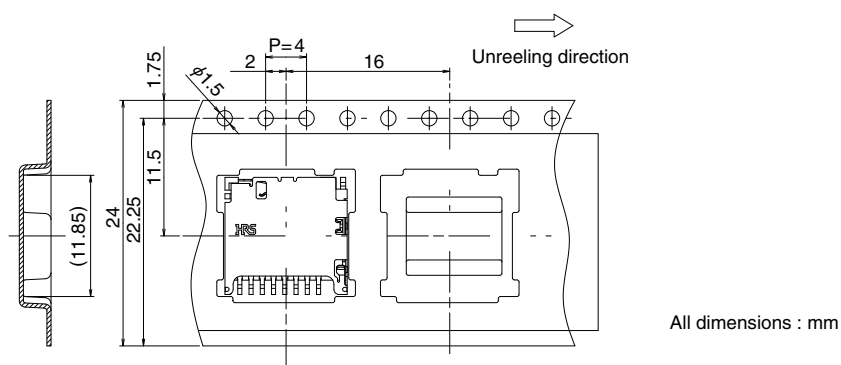
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Card detection switch	
Without the card	Card inserted
Open  (A) (B)	Closed  (A) (B)

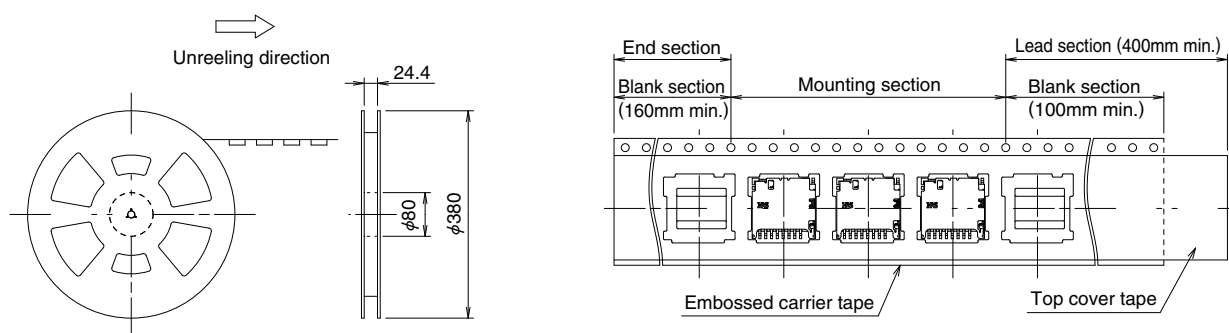
All dimensions : mm

◆ Packaging Specifications

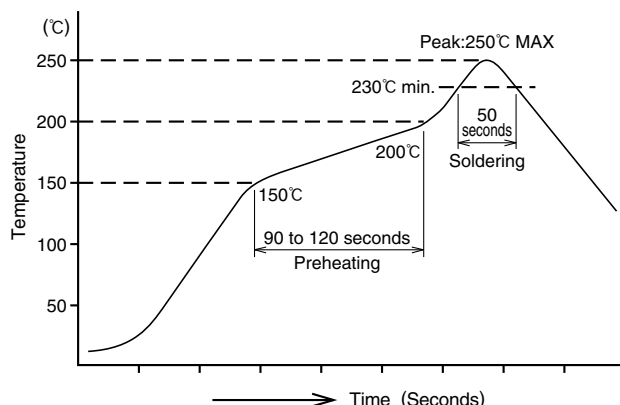
● Embossed carrier tape dimensions (2,000 pcs/reel)



● Reel Dimensions



◆ Recommended temperature profile



HRS test condition

Solder method : Reflow, IR/hot air
 Environment : Room air
 Solder composition : Paste, 96.5%Sn/3.0%Ag/0.5%Cu
 (Senju Metal Industry, Co., Ltd.'s
 Part Number:M705-GRN360-K2-V)
 Test board : Glass epoxy 60mm×100mm×1.0mm thick
 Metal mask : 0.12mm thick
 Number of reflow cycles : 2cycles max.

The temperature profiles shown are based on the above conditions.

In individual applications the actual temperature may vary, depending on solder paste type, volume / thickness and board size / thickness. Consult your solder paste and equipment manufacturer for specific recommendations.

◆ Precautions

1. Do not immerse or clean the entire connector with cleaning solutions as this may affect proper operation of the ejection mechanism and electrical performance of the connector
2. Do not apply excessive force to the connector when handling or after installation on the PC board.
3. The connectors will reliably connect and operate with the correctly inserted microSD™ cards.
Follow the correct insertion / ejection procedure for the specific connector in use.
Attempts of incorrect insertion of the card may cause damage to the connector or the card.
4. The connector must be correctly mounted on the PC board before the card can be inserted. Do not insert the card in the un-mounted connector.
5. Mounting on the Flexible Printed Circuit (FPC)
To assure correct performance it is recommended that a flat reinforcement plate 0.3 mm min. thick be used under the FPC.
6. Small visible residual manufacturing fluids or tooling marks do not affect connector's performance.
7. Repeated insertions and removal of the cards may leave some marks on the card itself. This will have no affect on the connector performance.

● Refer to applicable Operation Manual listed below for additional precautions.

Series	Operation Manual Number
DM3AT Series	ETAD-F0345
DM3BT Series	ETAD-F0324
DM3CS Series	ETAD-F0335
DM3D Series	ETAD-F0353

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